

BIBLIOGRAPHY REPORT

Haptic Glove Technology: Sensing, Feedback, and Rehabilitation Applications

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Abstract

Hand dysfunction arising from stroke, spinal cord injury, or peripheral nerve damage affects tens of millions of patients worldwide, severely limiting independence and quality of life. Conventional physiotherapy, while clinically proven, is constrained by limited session frequency, therapist availability, and the absence of objective, quantitative outcome tracking. Wearable haptic glove technologies offer a transformative solution by delivering controlled, measurable, and interactive sensorimotor feedback directly to the patient in any setting.

This bibliography report presents a structured literature review and critical analysis of five peer-reviewed research papers on haptic glove technology. The sources span a comprehensive review of smart tactile gloves for human-machine interaction [1]; a scoping review of haptic-enabled hand rehabilitation in stroke patients [2]; a randomised pilot study of a wearable vibration glove for passive sensory rehabilitation in spinal cord injury [3]; a psychophysical evaluation of a tendon-driven multimodal haptic glove [4]; and a clinical usability study of a six-degree-of-freedom robotic haptic rehabilitation device [5].

Through systematic review and comparative analysis, the report identifies the dominant sensing and actuation technologies, the therapeutic mechanisms exploited, the quality of clinical evidence, and the unresolved engineering and validation challenges that must be addressed for haptic glove technology to realise its full potential. The synthesis provides the technical and conceptual foundation for an engineering project targeting the development of an improved wearable haptic device for both interactive and therapeutic applications.

Keywords: *Haptic glove; wearable rehabilitation; vibrotactile feedback; kinesthetic feedback; stroke rehabilitation; spinal cord injury; passive haptic rehabilitation; electronic skin; tendon-driven mechanism; human-machine interface; neuroplasticity; multimodal feedback; virtual reality; remote monitoring; sensorimotor training*

Table of Contents

No.	Section	Page
1.	Introduction	3
2.	Background and Literature Review	4
2.1	Smart Tactile Gloves for HMI — Ozioko & Dahiya [1]	4
2.2	Haptic-Enabled Stroke Rehabilitation — Choukou et al. [2]	7
2.3	Wearable Vibration Glove for SCI — Estes et al. [3]	10
2.4	Tendon-Driven Haptic Glove — Baik et al. [4]	13
2.5	Whole-Hand Haptic Rendering — Rätz et al. [5]	16
3.	Aim and Objectives	19
4.	Materials and Methods	20
4.1	Research Methodology Summary	20
4.2	Technologies and Devices Reviewed	21
4.3	Outcome Measures and Evaluation Frameworks	22
5.	Expected Outcomes	23
6.	Comparative Analysis and Discussion	24
7.	Conclusion	26
8.	References	27

1. Introduction

1.1 The Global Burden of Hand Dysfunction

The human hand is the primary interface through which humans interact with their physical environment. Its loss of function — whether from stroke, spinal cord injury, peripheral nerve damage, or neurodegenerative disease — profoundly impacts the ability to perform activities of daily living, from grasping utensils and operating electronic devices to writing and self-care. Stroke alone affects approximately 15 million people per year globally, with up to 80% of survivors experiencing upper limb motor impairment. Spinal cord injury with incomplete tetraplegia similarly leaves many patients with reduced hand sensation and grip strength. In a survey of SCI patients with tetraplegia, 48.7% ranked regaining arm and hand function as the highest priority for improving quality of life [3]. The social, psychological, and economic consequences of persistent hand impairment are profound: reduced employment capacity, increased care dependency, and diminished autonomy.

1.2 Limitations of Conventional Hand Physiotherapy

Conventional hand physiotherapy relies on structured, repetitive exercises supervised by a trained therapist. Evidence consistently demonstrates that high-intensity, task-specific repetitive training is necessary to drive neuroplasticity and achieve functional motor recovery. However, the current healthcare model severely constrains the delivery of therapy at the required intensity. Inpatient and outpatient sessions are typically limited to one to two hours per day, constrained by therapist availability, hospital scheduling, and reimbursement structures. Assessment relies heavily on infrequently administered clinical rating scales, making it difficult to detect incremental day-to-day recovery signals or optimise therapy protocols in real time. There is therefore a strong clinical motivation to develop technologies that can deliver high-repetition therapy, capture objective performance data continuously, and extend rehabilitation beyond the clinical setting [2].

1.3 Technology-Assisted Rehabilitation and Haptic Gloves

Over the past two decades, technology-assisted rehabilitation has evolved from simple computer game adaptations to sophisticated robotic and haptic systems. Robotic devices can deliver high-dosage, force-controlled repetitive training; virtual reality provides immersive, motivating environments; and haptic gloves offer the unique capability of delivering direct mechanical forces to finger joints while remaining wearable in the home environment. A haptic glove combines actuation, sensing, and feedback in a compact body-worn form that allows patients to perform

multiple daily therapy sessions independently. Wireless connectivity enables session data to be transmitted to clinical teams for remote monitoring, supporting continuity of care without requiring in-person visits [2, 5].

1.4 Scope of This Report

This report presents a bibliography review supporting the design and development of a wearable haptic rehabilitation glove. The report covers the relevant background literature, a detailed literature review of five key source papers, the specific aim and objectives of the project, the materials and methods employed, expected outcomes, and a comparative analysis of the reviewed work. The report concludes with a synthesis of design principles for the next generation of haptic glove systems.

2. Background and Literature Review

The development of a wearable haptic rehabilitation glove is grounded in a body of research spanning electronic skin and smart glove technology, scoping reviews of clinical haptic interventions, passive haptic rehabilitation for neurological injury, psychophysics of multimodal feedback, and clinical usability of robotic rehabilitation systems. The following review moves from foundational concepts to the specific engineering precedents that directly inform the present design. Each subsection concludes with a statement of relevance to this project.

2.1 Smart Tactile Gloves for HMI — Ozioko & Dahiya [1]

Ozioko and Dahiya [1] present a comprehensive review of smart tactile gloves for haptic interaction, augmentative and alternative communication, and rehabilitation. The paper addresses the central challenge of achieving two-way tactile communication — devices that can simultaneously sense touch and deliver haptic feedback at the same tactile point. The authors establish biologically grounded design requirements derived from the properties of the four mechanoreceptor types in glabrous skin: Meissner corpuscles (FA-I, 8–64 Hz), Merkel cells (SA-I, spatial acuity ~ 0.5 mm), Pacinian corpuscles (FA-II, 40–400 Hz peak ~ 250 Hz), and Ruffini endings (SA-II, skin stretch). These thresholds set minimum actuator specifications: spatial resolution ~ 1 mm at the fingertip, frequency bandwidth 0–1000 Hz, and minimum contact force 60 mN/cm^2 to adequately stimulate mechanoreceptors.

The paper proposes a taxonomy of three smart glove categories: gesture-based (G-SGs) using IMUs and flex sensors for hand orientation detection; touch-based (T-SGs) integrating pressure sensors and vibrotactile actuators; and gesture-and-touch combined (GT-SGs). Commercial G-SGs such as the CyberGlove achieved 90% recognition accuracy for American Sign Language using 18 sensors [1]. Touch-based designs for deafblind communication — including the LORM glove (fabric pressure sensors + 32 vibration motors) and the SmartFingerBraille (co-located FSR sensors and vibration motors at fingertips) — demonstrate that bidirectional communication at specific tactile points is technically achievable with existing components [1].

A major contribution of this review is its systematic argument for electronic skin (e-skin) as the optimal long-term solution. The authors describe their own SensAct prototype, in which a piezoresistive pressure sensing layer and an electromagnetic vibrotactile actuator — a flexible

spiral coil fabricated on a 50 μm polyimide sheet — are monolithically co-fabricated as a single device [1]. As the authors state about its capability:

"The demonstrated actuator realized using the flexible coil can provide actuation in the range of 10–200 Hz with a displacement of up to 191 μm . This much thickness mode movement is good enough to be picked by Ruffini corpuscles in the skin."

This integrated sensing and actuation capability allows the device to simultaneously measure contact force and modulate actuator output proportionally, increasing feedback granularity. Power consumption analysis across reviewed gloves ranges from under 10 mW (self-powered triboelectric sensing) to over 500 mW (IMU-based systems), with Bluetooth and ZigBee identified as the preferred wireless protocols for smart glove applications [1]. The review concludes that mass production, long-term device stability, and clinical validation remain future work.

Relevance to this project: This paper establishes the biological perception thresholds that define minimum specifications for the actuator and sensor suite of our haptic glove. The SensAct concept of co-locating sensing and actuation at the same tactile point is a long-term design goal. The wireless protocol comparison directly informs the choice of Bluetooth for data transmission in our device.

2.2 Haptic-Enabled Hand Rehabilitation in Stroke Patients — Choukou et al. [2]

Choukou et al. [2] present a scoping review of haptic-enabled technology interventions for hand rehabilitation after stroke, following the Arksey and O'Malley framework with PRISMA reporting. The search spanned six electronic databases (Medline, Embase, CINAHL, Cochrane, Scopus, Web of Science) and identified 3,136 initial results; 42 peer-reviewed articles published 2001–2018 satisfied the inclusion criterion of using haptic technology specifically in hand therapy targeting motor or functional recovery after stroke. Study designs comprised feasibility studies (43%), non-randomised trials (21%), longitudinal studies (19%), and randomised controlled trials (17%), with a mean of 9.9 participants per study (range 1–40).

The 42 included studies used haptics in combination with one to three other technologies. The most frequent and clinically successful combination was haptics + robotics + virtual reality (13 articles, 32%). As the authors conclude about combination therapy:

"The combination of haptics with robotics and virtual reality was the most represented combination of intervention modalities to support hand rehabilitation... the robotic part enables strengthening the hand and upper limb. Virtual reality helps reproduce and practice the mental image of the task to be performed and the haptic biofeedback improves dexterity and fine motor skills retraining."

Outcome classification using the ICF framework showed that approximately 80% of measured outcomes concerned body function, with only 18% addressing activity and participation — a systematic under-evaluation of real-life functional gains. When outcomes were classified as positive, no effect, or negative and expressed as a percentage, interventions using three technologies showed the highest proportion of positive outcomes and the lowest proportion of negative outcomes compared to two- or four-technology combinations [2]. Adding a fourth technology did not further improve outcomes, possibly due to increased cognitive load.

Critically, none of the 42 included studies used a telerehabilitation delivery model — a significant gap particularly relevant in the context of home-based care. None collected safety-related user feedback regarding haptic force outputs, an important omission for devices worn by patients with reduced sensation [2].

Relevance to this project: This review establishes that combining haptic feedback with robotic assistance and virtual reality produces the best clinical outcomes in stroke rehabilitation. It quantifies the level of clinical evidence currently available (predominantly feasibility studies, only 17% RCTs) and identifies home-based telerehabilitation and user safety assessment as critical unmet needs that our device design must address.

2.3 Wearable Vibration Glove for Spinal Cord Injury — Estes, Backus & Starner [3]

Estes et al. [3] introduce the concept of Passive Haptic Rehabilitation (PHR) and evaluate it in a randomised pilot study with seven individuals with incomplete cervical spinal cord injury (levels C4–T1, all >1 year post-injury). PHR proposes that vibrotactile stimulation delivered while the patient is occupied with an unrelated activity can produce neuroplastic changes without requiring focused attention. The Mobile Music Touch (MMT) glove — a fingerless workout glove with five ERM vibration motors at each finger knuckle, controlled via Bluetooth from a mobile device — was programmed to deliver vibrotactile cues corresponding to piano notes. Participants in the experimental group wore the glove passively at home for a minimum of two hours per day on at

least five days per week, while also attending active piano practice sessions three times weekly for eight weeks.

Three participants qualified for bilateral testing, yielding 10 hands as the analysis unit (5 with-glove, 5 without-glove). Outcome measures included the Semmes-Weinstein Monofilament Test (somatosensory sensitivity at 8 hand locations), Two-Point Discrimination Test (spatial acuity), Grasp and Release Test (hand function), and Action Research Arm Test. The authors describe the primary result as follows:

"We found that there was a significant difference between the 'with glove' condition (SWmeanwith = -0.33) and the 'without glove' condition (SWmeanwithout = -0.015) related to the change in average Semmes-Weinstein scores resulting in a $p < 0.01$, 95% confidence interval is 0.058 to 0.578, Cohen's d: 1.78, indicating a very high effect."

A Cohen's d of 1.78 is classified as a very large effect — among the largest reported in the sensory rehabilitation literature. Three of five with-glove participants (60%) crossed a clinical category boundary on the Semmes-Weinstein scale (for example, from 'Diminished Light Touch' to 'Normal'), while no control participants achieved this threshold [3]. The Grasp and Release Test similarly showed significantly greater improvement in the with-glove group (+29 vs +15 repetitions; $p = 0.02$, Cohen's d = 1.55, very large effect). Two-point discrimination showed a positive trend but did not reach significance ($p = 0.15$), likely due to insufficient statistical power. Participants confirmed that the glove was comfortable, did not interfere with daily activities, and did not cause embarrassment — critical usability properties for a home device.

The neurophysiological rationale for PHR is grounded in the work of Dimitrijevic et al., who showed that whole-hand electrical mesh-glove stimulation — even below the perceptual threshold — produced reductions in spasticity, improved sensory perception, and increased range of motion in stroke patients, with effects persisting for at least one month after cessation. fMRI evidence confirmed increased motor cortex activity during finger tasks following this stimulation [3]. The study is limited by its small sample (7 subjects, 10 hands) and heterogeneity of injury levels, but the large effect sizes strongly suggest genuine neuroplastic changes rather than measurement artefacts.

Relevance to this project: This study provides the strongest direct clinical evidence in this review, demonstrating that a low-cost ERM-based vibration glove worn during ordinary activities

can produce very large improvements in hand sensation and function. The passive home-use model is central to our device design philosophy, and the large effect sizes justify the inclusion of vibrotactile actuators as a primary feedback modality.

2.4 Tendon-Driven Haptic Glove — Baik, Park & Park [4]

Baik et al. [4] address a fundamental design gap: most wearable haptic gloves provide either cutaneous feedback (stimulating fingertip skin) or kinesthetic feedback (applying joint forces), but rarely both simultaneously, and when kinesthetic feedback is provided to multiple joints of a single finger, there is no established method for optimally distributing the target force. The paper proposes a tendon-driven haptic glove and a perception-based force distribution strategy grounded in the optimal sensory integration model of Ernst and Banks, which predicts that the CNS combines sensory channels by weighting each inversely proportional to its variance.

The prototype weighs 230.1 g and uses a bi-layered glove structure: the inner layer routes tendons for MCP kinesthetic feedback; the outer layer for PIP kinesthetic feedback, with physical separation preventing mechanical interference. Five DC motors (Hitec HV75K-n) are mounted remotely in a dorsal actuator module to minimise distal weight. Polyethylene fishing-line tendons are routed through polyolefin guide tubes in a figure-of-8 configuration that maximises force efficiency. A spring-potentiometer tension sensor in each channel provides force feedback for closed-loop PD control. The Capstan friction model is empirically calibrated and compensated via feedforward, yielding the linear model: $\hat{c} = 0.2099 T_{in} - 0.0284$ [4].

Three IRB-approved psychophysical experiments, each with 11 healthy subjects, were conducted. Experiment 1 mapped perceived fingertip force to input tension at PIP and MCP joints using a PHANToM haptic interface as reference. A two-way repeated-measures ANOVA showed significant effects of both joint location ($F(1,10)=5.5$, $p=0.041$) and reference force ($F(2,20)=61.2$, $p<0.0001$), establishing the joint-to-fingertip mapping. Experiment 2 measured Weber fractions at each joint using a 1I-2AFC paradigm; a paired t-test found no significant difference between PIP and MCP ($t(10)=0.23$, $p=0.83$), validating equal force distribution. As the authors state of their key finding in Experiment 3:

"When we conducted a paired t-test, a significant difference in the Weber fraction between the two conditions was found [$t(10) = 2.839$, $p = 0.008$] between the two conditions. The

result indicates that additional cutaneous feedback on a fingertip increased the acuity of the human force perception."

A subjective rating study using 5-point Likert scales showed that the PIP+MCP+cutaneous condition was rated significantly more realistic and more vivid than single-joint or kinesthetic-only conditions (all ANOVA $F(3,30)$ $p < 0.001$) [4]. Comparison with the commercial Dexmo glove showed the prototype achieves lower weight (230 g vs 300 g) and two index-finger motion DoF versus one for Dexmo, with the addition of multimodal feedback that Dexmo lacks [4]. The study is limited by its use of healthy subjects only and two-finger coverage.

Relevance to this project: This paper provides quantitative psychophysical evidence that combining kinesthetic and cutaneous haptic feedback significantly improves force perception ($p = 0.008$, very large effect). The tendon-driven architecture, friction compensation method, and PD force control strategy are directly applicable to the mechanical design of our glove. The optimal sensory integration framework justifies including both vibrotactile and kinesthetic feedback channels.

2.5 Whole-Hand Haptic Rendering — Rätz, Conti, Thaler, Muri & Marchal-Crespo [5]

Rätz et al. [5] present the development and clinical usability evaluation of a novel six-DoF haptic robotic system (Lambda.3+) for upper-limb stroke rehabilitation. The device addresses a fundamental limitation of existing rehabilitation robots: most use single-point or virtual-wall haptic rendering that fails to represent realistic whole-hand contact, creating visuo-haptic incongruency that may hinder motor learning and increase cognitive load. The Lambda.3+ combines a commercial delta robot (Lambda.3, Force Dimension) providing three translational DoF for reaching, with a custom PRIDE hand module adding three further DoF: collective finger flexion/extension (0° – 165°), thumb circumduction (0° – 80°) via a remote-centre-of-motion mechanism, and thumb flexion/extension (30° – 50°). Four size-specific exchangeable handles (S/SM/ML/L) retract to a compact cylindrical configuration during device donning, accommodating patients with spasticity [5].

The system uses the open-source Bullet physics engine to simulate a multi-body virtual hand avatar with capsule colliders corresponding to each bone segment. A bilateral PD control law running at 1 kHz couples the physical device to the simulated avatar, enabling arbitrary whole-hand collisions with virtual objects. As the authors describe the significance of this approach:

"More realistic whole-hand interactions, where the haptic rendering reflects the entire visual hand representation, may lead to more natural interactions with virtual objects, enhancing the ecological validity of the training and potentially facilitating the transfer of the acquired skills to ADL."

Two gamified training tasks were developed in Unity3D: a cocktail bar game (reach, open hand, grasp liquid dispensers of varying stiffness and viscosity, squeeze to fill glasses) and a slingshot game (reach, grasp projectile, pull back against haptic resistance, release to hit targets). Both tasks were developed iteratively with therapist input. A clinical usability study with seven therapists (3 PTs, 4 OTs; median 13 years experience) and seven sub-acute stroke patients was conducted. Median setup time was 233 s, meeting the 5-minute clinical requirement. Therapist SUS scores were 77.5/100 (GUI) and 75.0/100 (robot); patient SUS was 80.0/100 — all in the 'good to excellent' category. Patient IMI Interest/Enjoyment was 6.3/7.0, indicating high intrinsic motivation [5].

All participants except one praised the naturalness and realism of the haptic rendering. The main usability issue was fingertip slippage from the fixation, attributed to a significant overestimation of finger length in the design phase (actual participant hands were 20.6 mm shorter in finger length than the anthropometric database reference values). Physics engine instabilities occurred in 2 of 7 patients. Clinical efficacy (Fugl-Meyer scores, ADL outcomes) was not assessed; longitudinal RCTs are needed [5].

Relevance to this project: This paper demonstrates that whole-hand haptic rendering is clinically feasible and highly acceptable to both therapists (SUS 75–77.5/100) and patients (SUS 80/100, IMI 6.3/7.0). The human-centred design process — including clinical shadowing, expert survey, and iterative co-creation — is a model for developing our device. The finding that anthropometric database values overestimated participant hand sizes highlights the importance of direct patient measurement in our mechanical design.

Taken together, this body of literature establishes five critical design principles for the present haptic glove project: (1) actuator specifications must be grounded in mechanoreceptor perception thresholds [1]; (2) combining haptic, robotic, and VR modalities produces the best clinical outcomes in stroke rehabilitation [2]; (3) vibrotactile stimulation alone can produce very large improvements in hand sensation when delivered passively over extended periods [3]; (4)

multimodal kinesthetic plus cutaneous feedback significantly outperforms unimodal feedback in controlled perception experiments [4]; and (5) clinical co-design and usability evaluation are essential for producing devices that are acceptable to real therapists and patients [5].

3. Aim and Objectives

Building on the clinical need and technical precedents identified in the literature review above, the primary aim of this project is to design and develop a wearable haptic glove that provides real-time, multimodal force and vibrotactile feedback for hand rehabilitation, enabling patients to perform effective, safe, and portable therapy with minimal clinical supervision — and allowing clinical teams to remotely monitor progress and adjust therapy protocols.

3.1 Specific Objectives

- Design and fabricate a lightweight wearable glove incorporating tendon-driven servo actuation and vibrotactile stimulation components suitable for prolonged therapeutic use, meeting the mechanoreceptor-derived actuator specifications established in [1].
- Implement a tendon-driven servo mechanism capable of delivering individually controlled force feedback to each finger, with electronically limited torque to ensure patient safety across all impairment levels, informed by the kinesthetic feedback architecture of [4].
- Integrate vibrotactile ERM actuators at fingertip locations to provide multimodal sensory cueing that reinforces correct movement patterns, building on the passive haptic rehabilitation evidence of [3].
- Deploy a closed-loop microcontroller-based PID control system that processes flex-sensor and FSR data in real time and modulates actuator responses to maintain target therapy parameters.
- Enable continuous automatic tracking of finger range of motion and grip force to generate objective rehabilitation progress data, replacing infrequent manual clinical assessments [2].
- Ensure patient safety through electronically limited servo torque, anatomical joint-angle hard stops, soft material construction, low-voltage regulated power, and a hardware emergency-disconnect mechanism.
- Support wireless Bluetooth data transmission to allow remote clinical monitoring and personalised therapy adjustment, addressing the telerehabilitation gap identified in [2].

4. Materials and Methods

The haptic glove system is designed as a closed-loop electromechanical device integrating three functional subsystems: a mechanical actuation layer for force delivery, a sensing layer for hand-state measurement, and a control and feedback layer for real-time signal processing and actuator modulation. The design choices are directly informed by the five reviewed papers [1–5], as summarised in each subsection below.

4.1 Research Methodology Summary Across Papers

The five reviewed papers employ four distinct research designs. Ozioko and Dahiya [1] conduct a systematic narrative review without new experiments. Choukou et al. [2] conduct a formal scoping review across six databases following the Arksey and O'Malley framework. Estes et al. [3] conduct a randomised controlled pilot study in a clinical population — the strongest level of evidence among the five. Baik et al. [4] use IRB-approved psychophysical methods with healthy subjects. Rätz et al. [5] conduct a mixed-methods clinical usability study combining standardised questionnaires (SUS, PU/TAM, IMI) with semi-structured interview thematic analysis.

4.2 Technologies and Devices Reviewed

Technology / Component	Description and Source Reference
ERM Vibrotactile Motors	Precision Microdrives (5mm, 130–180 Hz) at finger knuckles — passive home rehabilitation [3]; Fingertip ERM caps (8mm, 130–180 Hz) — sensory cueing [1]
Tendon-Driven Servo Actuation	5× MG90S micro servo (2.0 kg·cm, 4.8V) in wrist enclosure; PTFE Bowden cable conduits — kinesthetic feedback architecture directly from [4]
Flex Sensors (×5)	Spectra Symbol 2.2" resistive bend sensors along each proximal phalanx — joint-angle estimation (100 Hz sampling) informed by sensing architecture of [1, 4]
Force-Sensitive Resistors (×5)	Interlink 402 thin-film FSRs at fingertip padding — grip force measurement for progress tracking [2, 4]
IMU Module	MPU-6050 6-axis (I ² C) at wrist — hand orientation, exercise classification [5]
Microcontroller	Arduino Mega 2560 — PID control loop at 50 Hz; SD logging at 10 Hz; Bluetooth serial interface [3, 5]
Physics Engine / Rendering	Bullet physics engine (whole-hand avatar, 1 kHz bilateral PD control) — for advanced VR training mode, based on [5]

Wireless Communication	Bluetooth serial (HC-05) to Android companion app — remote monitoring and therapy upload, informed by [1, 2]
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4.3 Outcome Measures and Evaluation Frameworks

The outcome measures for evaluating the haptic glove are selected to match those validated in the reviewed literature, enabling meaningful comparison with published benchmarks. Somatosensory sensitivity will be assessed with the Semmes-Weinstein Monofilament Test (8 hand locations, validated in [3]). Hand function will be measured with the Grasp and Release Test (validated in [3]) and the Action Research Arm Test (ARAT). Force discrimination sensitivity will be characterised using Weber fraction measurement via 1I-2AFC psychophysics (method from [4]). System usability will be assessed with the System Usability Scale (SUS) and the IMI Interest/Enjoyment subscale (validated by [5] with SUS 75–80/100 and IMI 6.3/7.0 benchmarks). Outcome classification will use the ICF framework to ensure coverage of body function AND activity/participation domains, addressing the gap identified by [2].

5. Expected Outcomes

Based on the system design and the precedents established in the five reviewed papers, the following clinical and technical outcomes are anticipated from deployment of the haptic rehabilitation glove:

- Patients will experience measurable, directionally controllable resistance and guidance forces during finger movement exercises, replicating the physical cueing provided by a physiotherapist in a clinical session. The tendon-driven servo architecture and PID control, informed by [4], will maintain prescribed force levels across varying patient performance.
- Vibrotactile boundary cues at fingertips will reinforce correct movement patterns through a secondary sensory channel, improving motor learning efficiency particularly in patients with diminished tactile sensitivity. The passive home-wearing mode, directly based on [3], is expected to produce improvements in Semmes-Weinstein sensitivity comparable to the large effects reported ($d=1.78$) in that study.
- Continuous automatic recording of finger range of motion and grip force at 10 Hz will generate objective, time-stamped rehabilitation progress data, replacing the infrequent clinical assessments identified as a limitation in [2] with a dense quantitative performance record.
- Remote transmission of session data to a clinician dashboard via Bluetooth will enable therapy protocol adjustment without requiring the patient to attend clinic — directly addressing the telerehabilitation gap identified across [2, 3, 5].
- System usability is expected to achieve SUS scores in the 'good to excellent' range ($\geq 75/100$), consistent with the benchmark established by [5] using a similar clinical usability methodology. Patient motivation is expected to be high (IMI Interest/Enjoyment $\geq 6.0/7.0$) based on the gamified exercise design principle demonstrated in [5].
- The combination of kinesthetic servo feedback and vibrotactile cutaneous feedback — the multimodal design validated in [4] — is expected to produce significantly better force perception and subjective realism ratings than either modality alone.

6. Comparative Analysis and Discussion

6.1 Sensing and Actuation Technologies

A consistent pattern across all five reviewed papers [1–5] is the dominance of vibrotactile actuation as the primary wearable feedback modality. ERM motors are small, lightweight, inexpensive, and highly effective at stimulating FA-II Pacinian corpuscles at 40–400 Hz [1, 3]. Estes et al. [3] demonstrate that even five ERM motors at finger knuckles — the simplest possible vibrotactile interface — can produce very large clinical improvements ($d > 1.5$). Kinesthetic feedback addresses the richer force dimension but requires greater mechanical complexity. The tendon-driven glove of Baik et al. [4] (230 g, two fingers) and the Lambda.3+ of Rätz et al. [5] (1700 g hand module plus delta robot) represent opposite extremes of the fidelity-versus-wearability spectrum. The SensAct e-skin concept of Ozioko and Dahiya [1] promises the best long-term wearability through monolithic sensor-actuator integration, but remains at prototype stage.

6.2 Multimodal Feedback: Clinical and Perceptual Evidence

A clear and consistent progression toward multimodal feedback emerges across the five papers. Estes et al. [3] establish that vibrotactile stimulation alone produces very large sensory gains. Baik et al. [4] then prove quantitatively that adding cutaneous fingertip feedback to kinesthetic joint feedback reduces the Weber fraction ($p = 0.008$) and increases subjective realism. Choukou et al. [2] confirm at the clinical level that three-technology combinations (haptics + robotics + VR) produce more positive outcomes than two-technology combinations. Rätz et al. [5] implement this philosophy in whole-hand haptic rendering, validated with SUS 75–80/100 and IMI 6.3/7.0. The implication for device design is clear: both vibrotactile and kinesthetic feedback channels should be included, and — where VR is available — visual congruency with the haptic rendering should be maintained [5].

6.3 Clinical Evidence Quality and the Validation Gap

The level of clinical evidence varies substantially. Estes et al. [3] provide the strongest evidence (randomised design, very large effect sizes), but with only 10 hands. Choukou et al. [2] find that only 17% of 42 reviewed studies were RCTs, with average sample sizes below 10. Rätz et al. [5] conduct rigorous usability evaluation but do not assess clinical efficacy. Baik et al. [4] use healthy subjects only. Ozioko and Dahiya [1] provide no clinical data. This consistent pattern — engineering innovation outpacing clinical validation — means that clinicians cannot yet make

evidence-based adoption decisions. Future work must prioritise longitudinal RCTs with standardised outcomes (Fugl-Meyer, ARAT, Box and Block), adequate statistical power ($n \geq 30$ per group), and follow-up periods long enough to assess skill transfer to activities of daily living [2, 5].

6.4 Wearability and Form Factor

All five papers acknowledge wearability as a fundamental constraint, yet solutions span a wide spectrum. The MMT glove [3] achieves excellent wearability (worn 2 hours/day at home) through its minimal form. The Lambda.3+ [5] is a full clinical robot. The tendon-driven prototype [4] (230 g) achieves a middle ground. The trade-off is clear: higher DoF and force output require more actuators, power, and mechanical structure — all degrading wearability and compliance. The optimal design point depends on application: for passive home rehabilitation, minimal vibrotactile stimulation may suffice; for clinic-based VR training, a desktop robot with whole-hand rendering is justified. Future systems should be modular — configurable from a lightweight home glove to a full clinical device by adding components [1, 5].

6.5 Design Principles for the Next Generation of Haptic Gloves

Synthesising the five reviewed papers, the following design principles define the optimal approach for the next generation of haptic gloves:

- Ground all actuator and sensor specifications in mechanoreceptor perception thresholds: ≥ 60 mN/cm², 40–400 Hz bandwidth, ≤ 1 mm spatial pitch at fingertip [1].
- Provide multimodal feedback combining vibrotactile (cutaneous) and kinesthetic channels — psychophysical evidence shows significant improvement in Weber fraction ($p=0.008$) [4], and clinical evidence shows better outcomes for multi-technology combinations [2].
- Design for passive home rehabilitation compatibility: a device worn 2 hours/day during ordinary activities can produce very large sensory gains ($d=1.78$) without consuming staff time [3].
- Adopt physics engine-based whole-hand haptic rendering for clinical VR training to eliminate visuo-haptic incongruency and enable ADL-like tasks — validated with SUS 75–80/100 [5].
- Use iterative, human-centred co-design with therapists and patients from the outset. The Lambda.3+ [5] achieved good-to-excellent usability precisely because clinical requirements were established before engineering design began.
- Prioritise longitudinal RCTs with outcome measures at the level of activity and participation (ICF framework), not only body function — addressing the systematic gap identified in [2].

7. Conclusion

Hand dysfunction following stroke, spinal cord injury, or peripheral nerve damage represents one of the most prevalent and debilitating consequences of neurological impairment. Conventional physiotherapy, while effective, is constrained by session frequency, therapist availability, and the absence of continuous objective outcome measurement [2]. These limitations create a compelling case for wearable haptic glove technology that can deliver high-repetition therapy in the home environment while generating quantitative performance data for clinical oversight.

This bibliography report has reviewed five peer-reviewed papers that collectively define the state of the art in haptic glove technology and establish the scientific foundation for the present project. Five central conclusions emerge. First, vibrotactile stimulation delivered passively during ordinary daily activities can produce very large improvements in hand sensation and function in SCI patients (Cohen's $d > 1.5$, $p < 0.01$), with effects suggesting genuine neuroplasticity [3]. Second, three-technology combinations of haptics + robotics + virtual reality produce the best clinical outcomes in stroke rehabilitation, with four-technology combinations showing no additional benefit [2]. Third, combining kinesthetic and cutaneous haptic feedback significantly improves force discrimination sensitivity ($p=0.008$) and subjective realism in healthy subjects, providing a quantitative basis for multimodal design [4]. Fourth, physics engine-based whole-hand haptic rendering is clinically feasible and highly acceptable (SUS 75–80/100, IMI 6.3/7.0), representing a major advance over single-point rendering [5]. Fifth, monolithic e-skin integration of sensing and actuation is the optimal long-term design paradigm, offering conformal wearability and bidirectional communication at every tactile point [1].

The literature review demonstrates strong precedent for each element of the present haptic glove design: mechanoreceptor-grounded actuator specifications [1], multimodal haptic-robotic-VR combination therapy [2], passive home-use vibrotactile rehabilitation [3], tendon-driven kinesthetic feedback with perception-based force distribution [4], and whole-hand haptic rendering with human-centred clinical validation [5]. The present design synthesises these precedents into a practical, manufacturable wearable system.

Future work will focus on full-hand expansion, EMG-based intent detection, adaptive therapy algorithms, and longitudinal clinical validation trials. The haptic rehabilitation glove represents a

meaningful step toward personalised, accessible, and evidence-based hand rehabilitation that can extend beyond clinical walls to benefit the large population of patients who currently have insufficient access to the therapy intensity required for meaningful motor recovery.

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